

*Custom Pipeline Monitoring **Framework For Distributed** **In-Memory** Batch Data Processing*

Yunata Gunawan - 14 November 2024

Tools

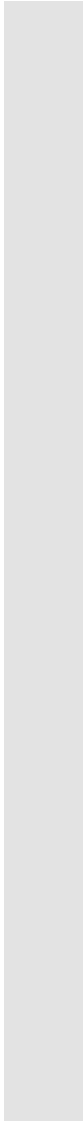
Data Pipeline



sbt

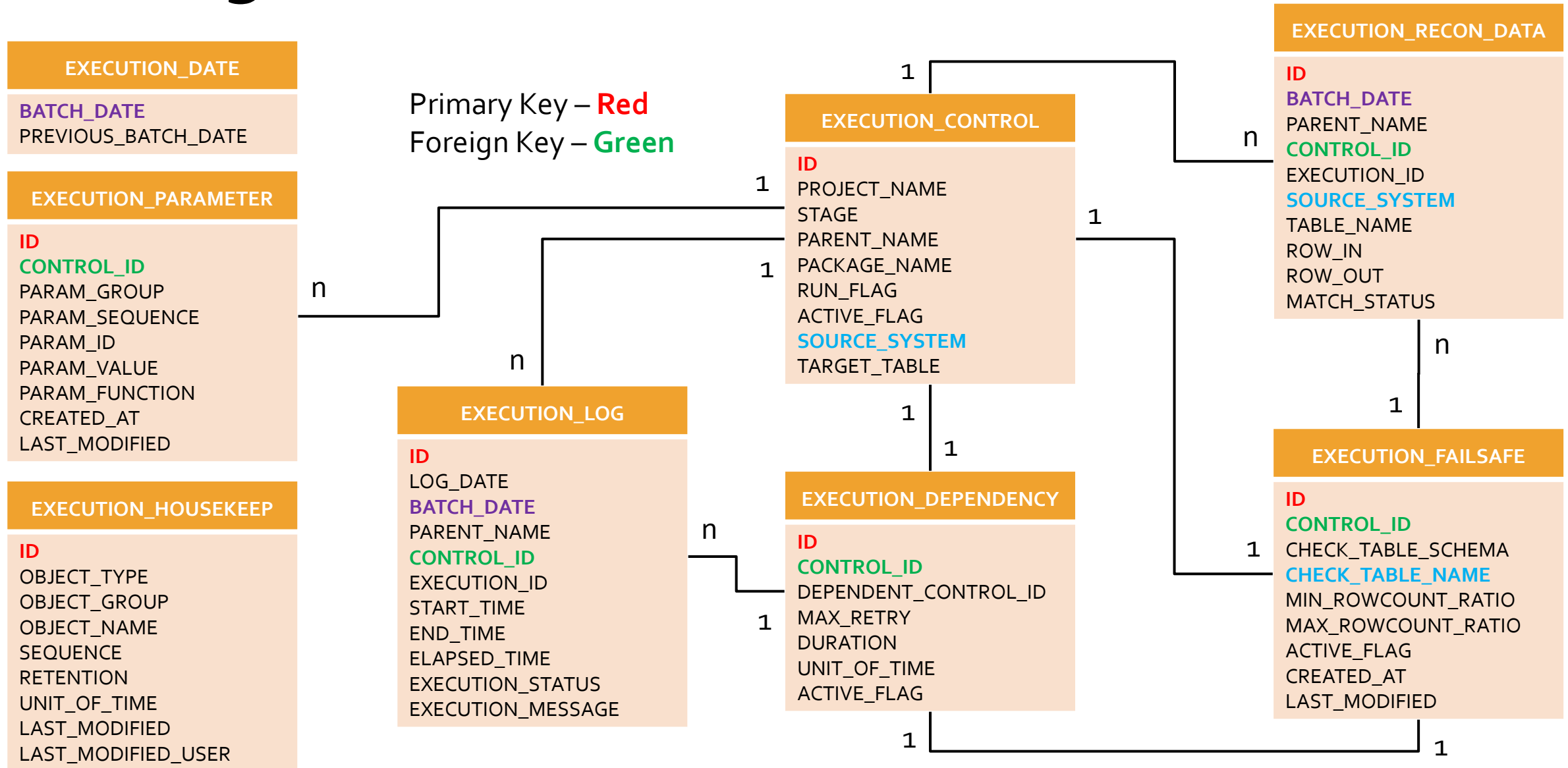
Database





Implementation

Management Framework ERD



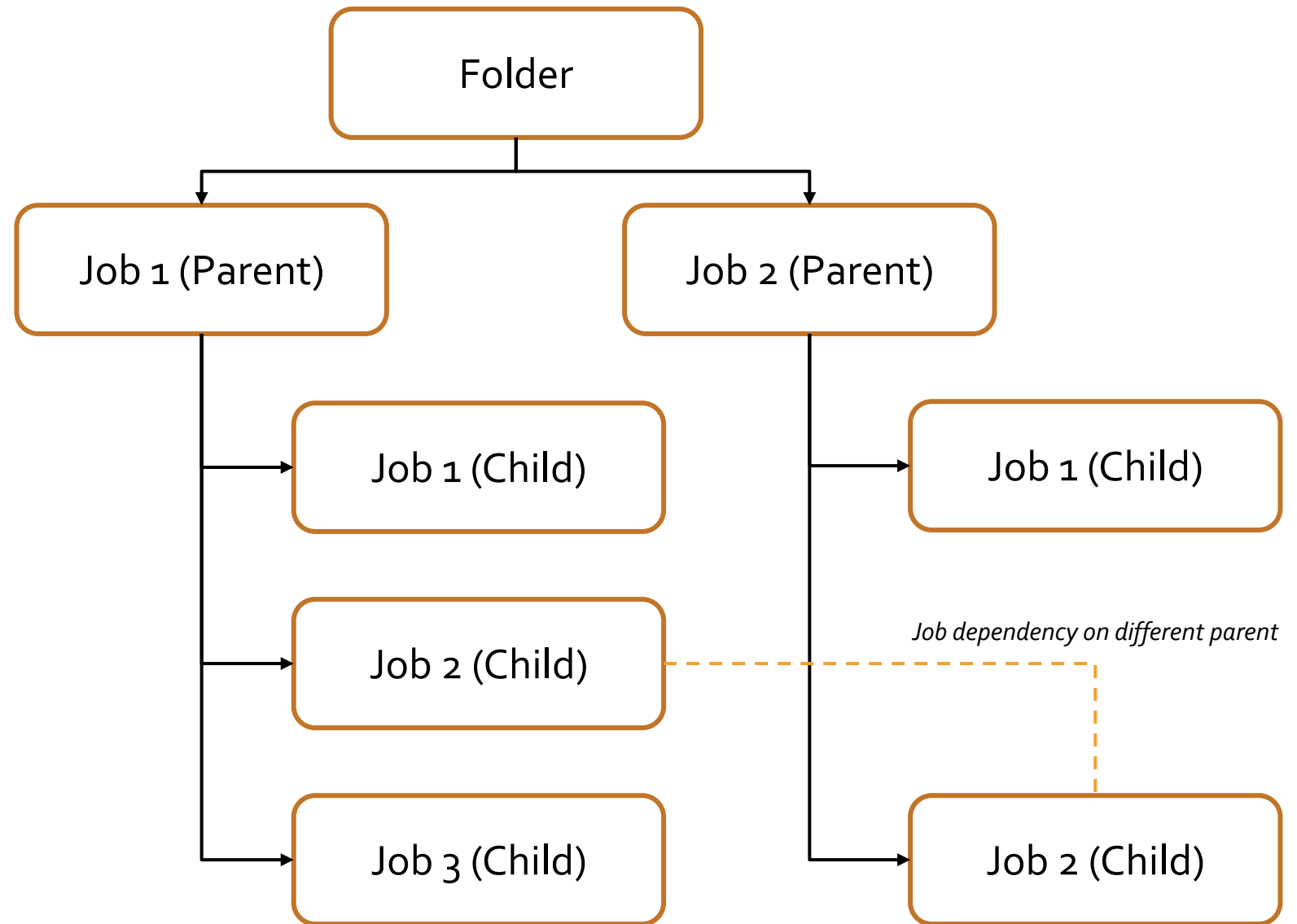
Management Framework Detail

Table Name	Purpose
EXECUTION_DATE	Store date of extraction, only 1 record
EXECUTION_CONTROL	List of ETL jobs, can running and is job active flag. Also include relation to parent and job
EXECUTION_HOUSEKEEP	List of housekeeping parameter, you can design your own data house keeping
EXECUTION_PARAMETER	List of processing parameter
EXECUTION_LOG	Log of running job and their status
EXECUTION_RECON_DATA	List of running job per date and their row count recon
EXECUTION_DEPENDENCY	List of job dependency on different parent

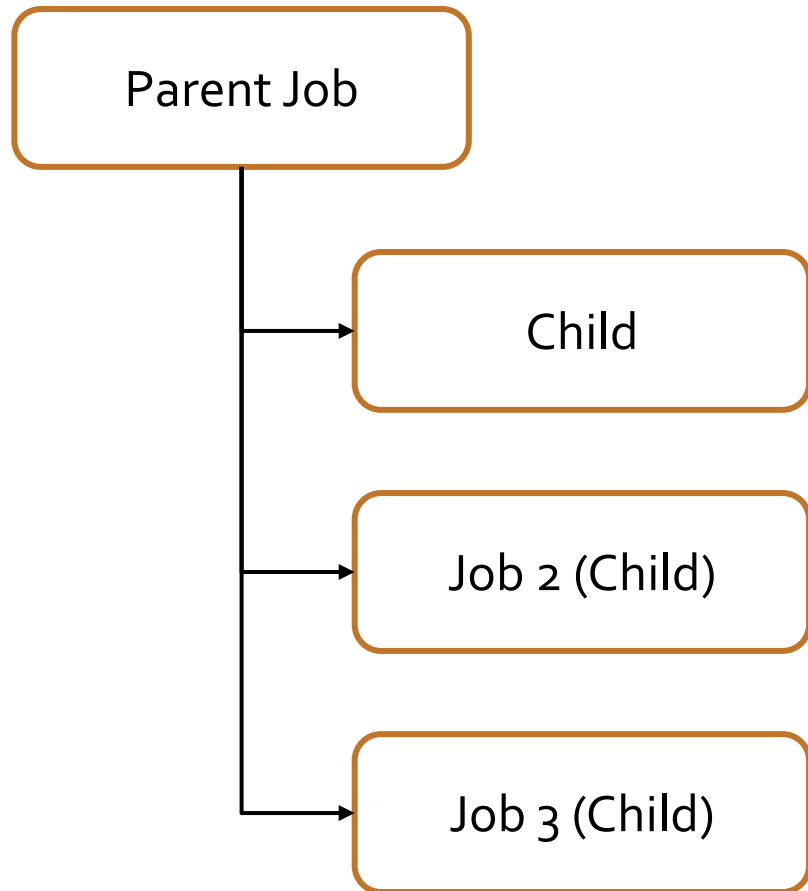
Unit of Time	Table Scope
SECOND	EXECUTION_DEPENDENCY
MINUTE	EXECUTION_DEPENDENCY
HOURL	EXECUTION_DEPENDENCY
DAY	EXECUTION_HOUSEKEEP

Job Status	Description
START	Job start but don't process data / initialization
PROCESS	Currently running to process data
SUCCESS	Stop running, no error
ERROR	Stop running, error

Spark Job Structure



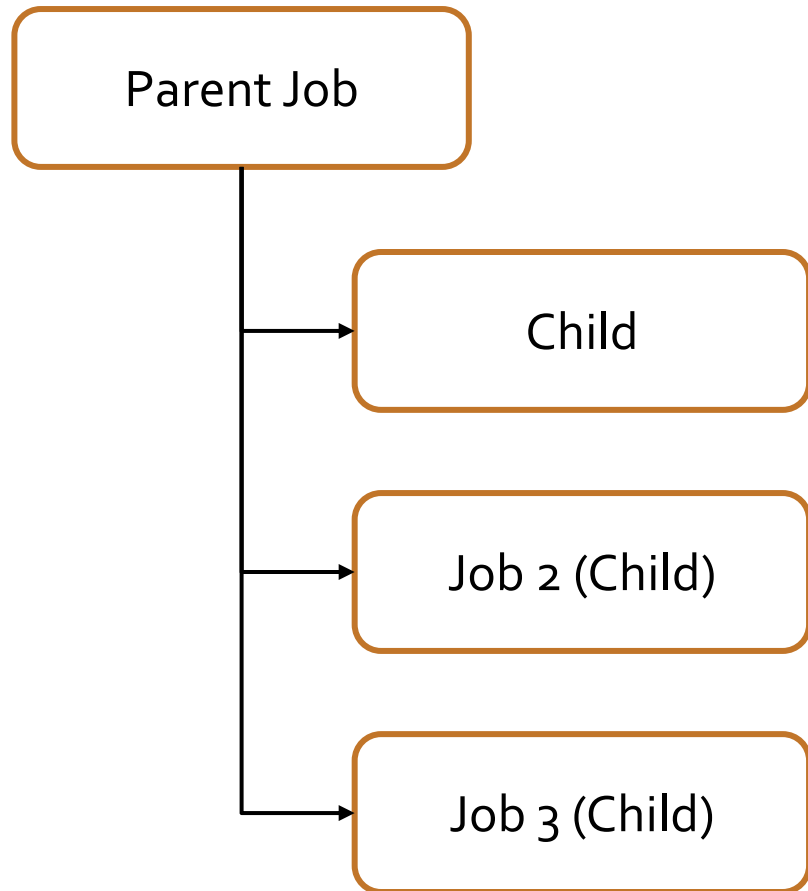
Execution Control



Job Name	RUN_FLAG	ACTIVE_FLAG
Parent Job	true	true
Child 1	true	true
Child 2	true	true
Child 3	true	true

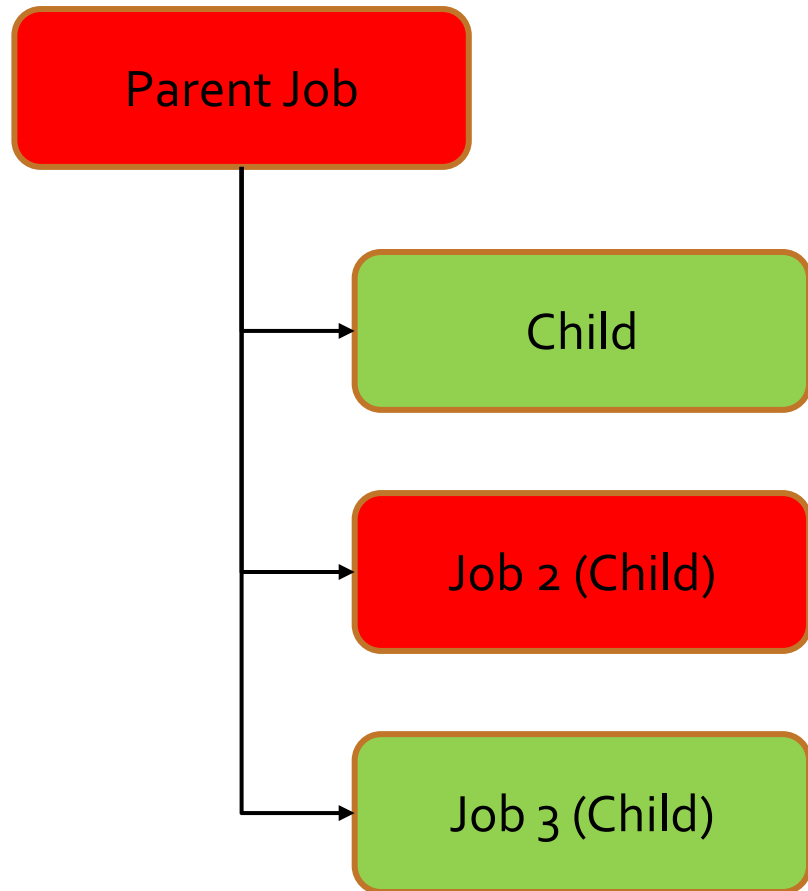
PS. You can switch off the job you didn't want to execute set the active flag to false.
EXECUTION_DEPENDENCY and ***EXECUTION_FAILSAFE*** also have this feature.

Execution Control: Success Run



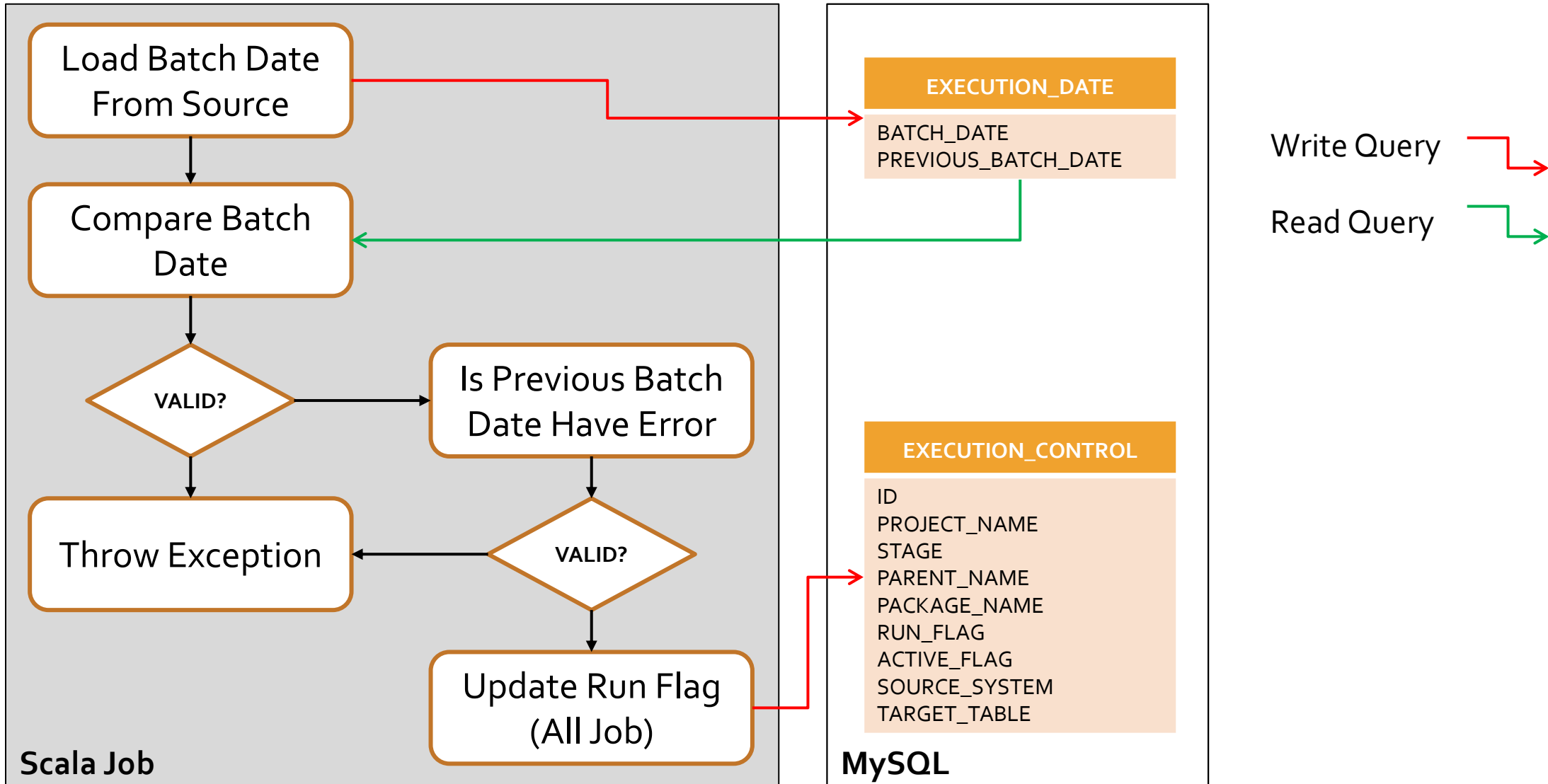
Job Name	RUN_FLAG	ACTIVE_FLAG
Parent Job	false	true
Child 1	false	true
Child 2	false	true
Child 3	false	true

Execution Control: Error

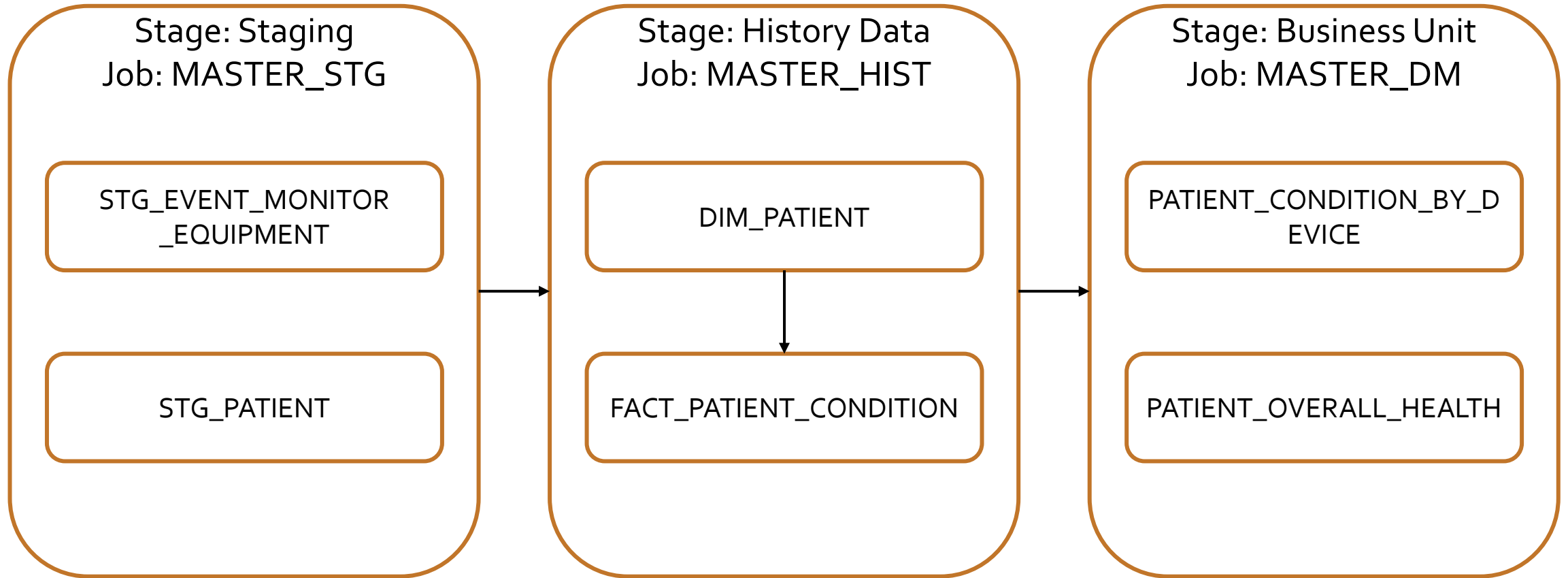


Job Name	RUN_FLAG	ACTIVE_FLAG
Parent Job	true	true
Child 1	false	true
Child 2	true	true
Child 3	false	true

Stage: Pre-ETL Master Job

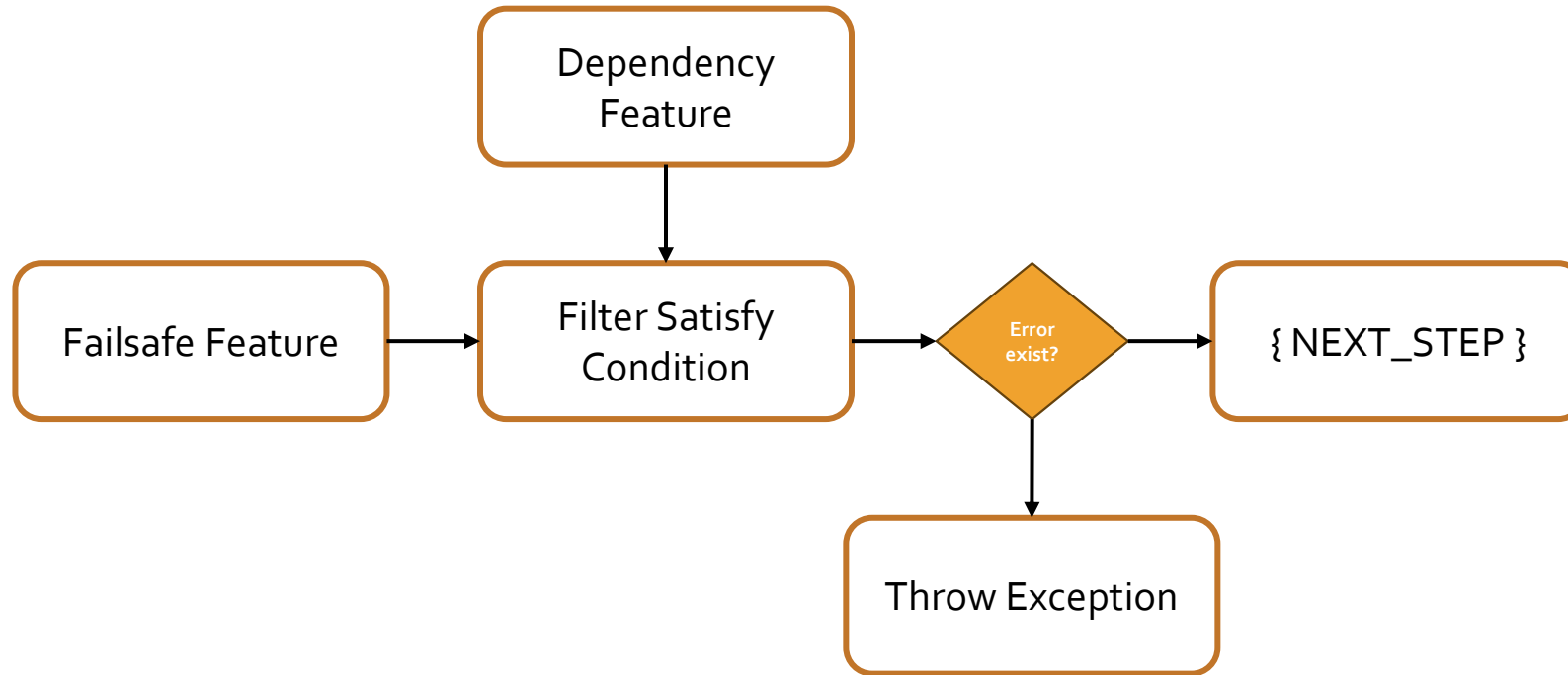


Spark Job Workflow

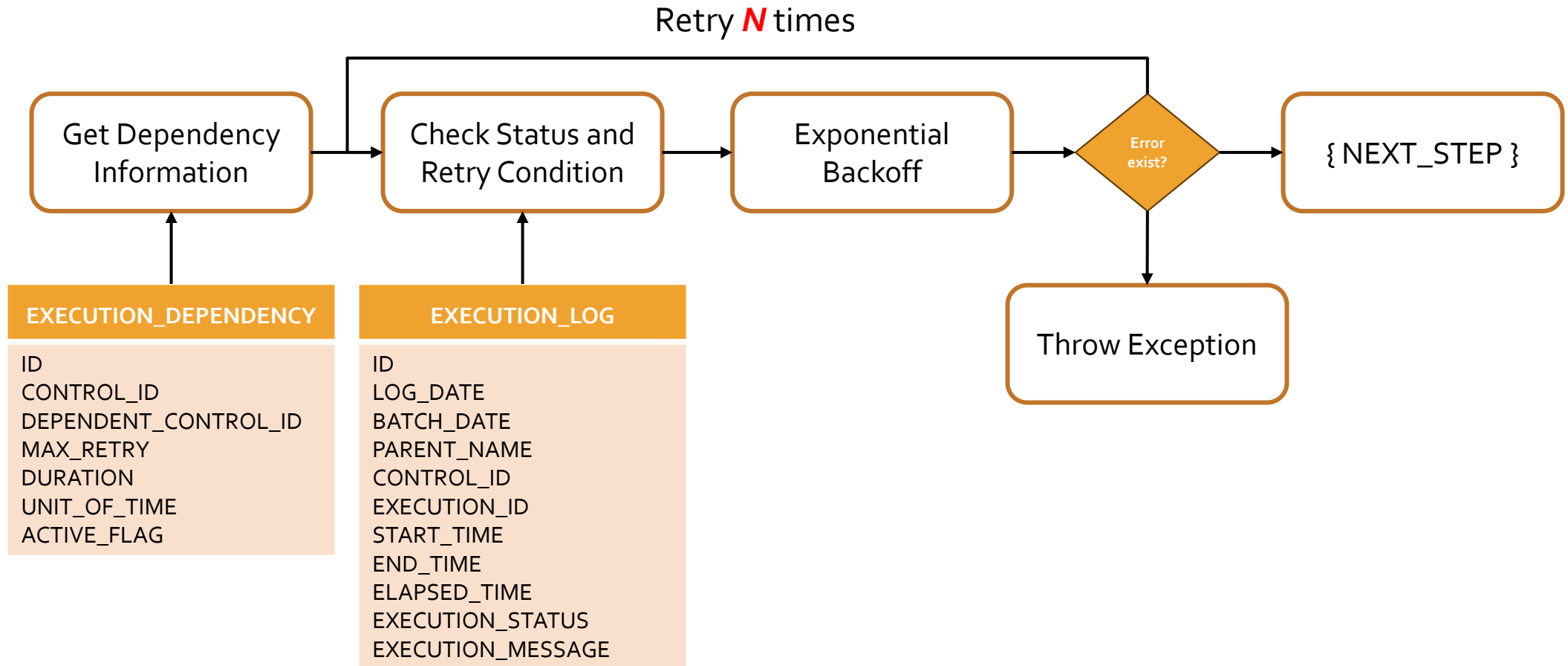


Arrow connecting 2 child or more meaning a sequential task, otherwise a parallel task

Dependency and Failsafe Features

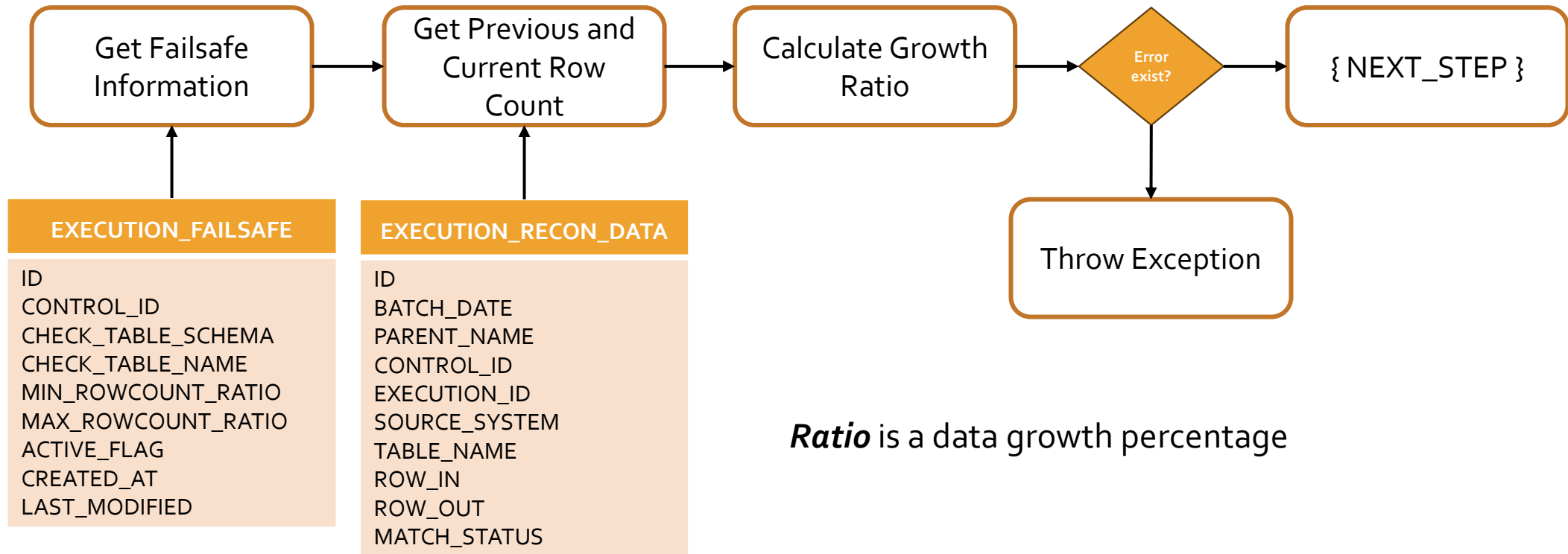


Dependency Feature

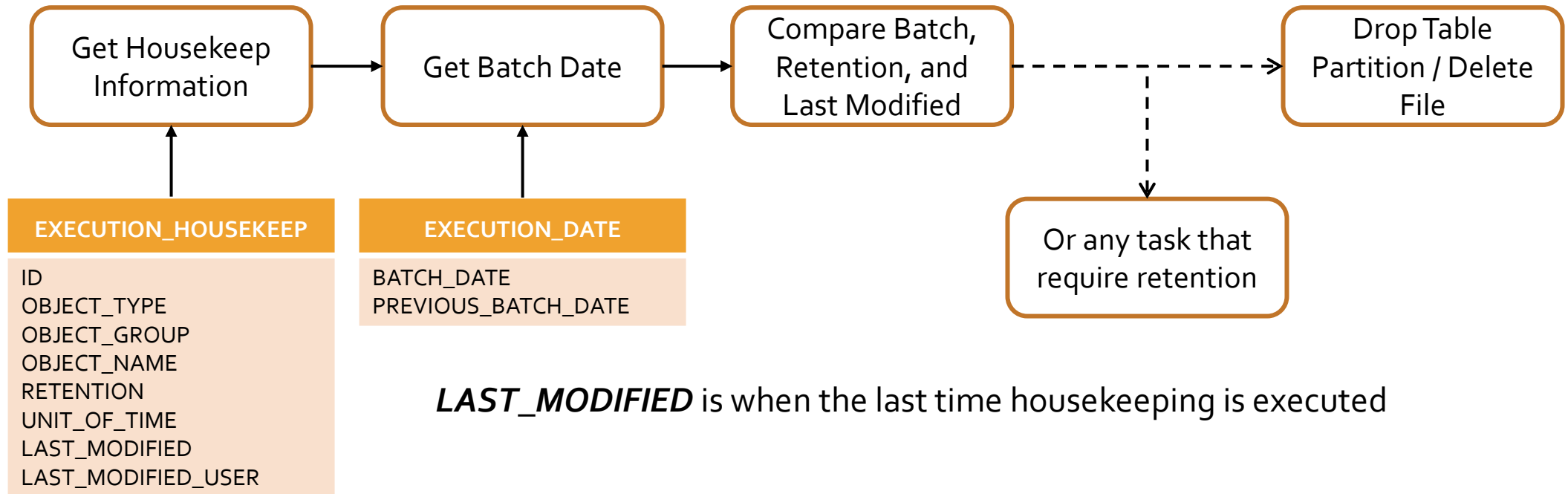


Failsafe Architecture

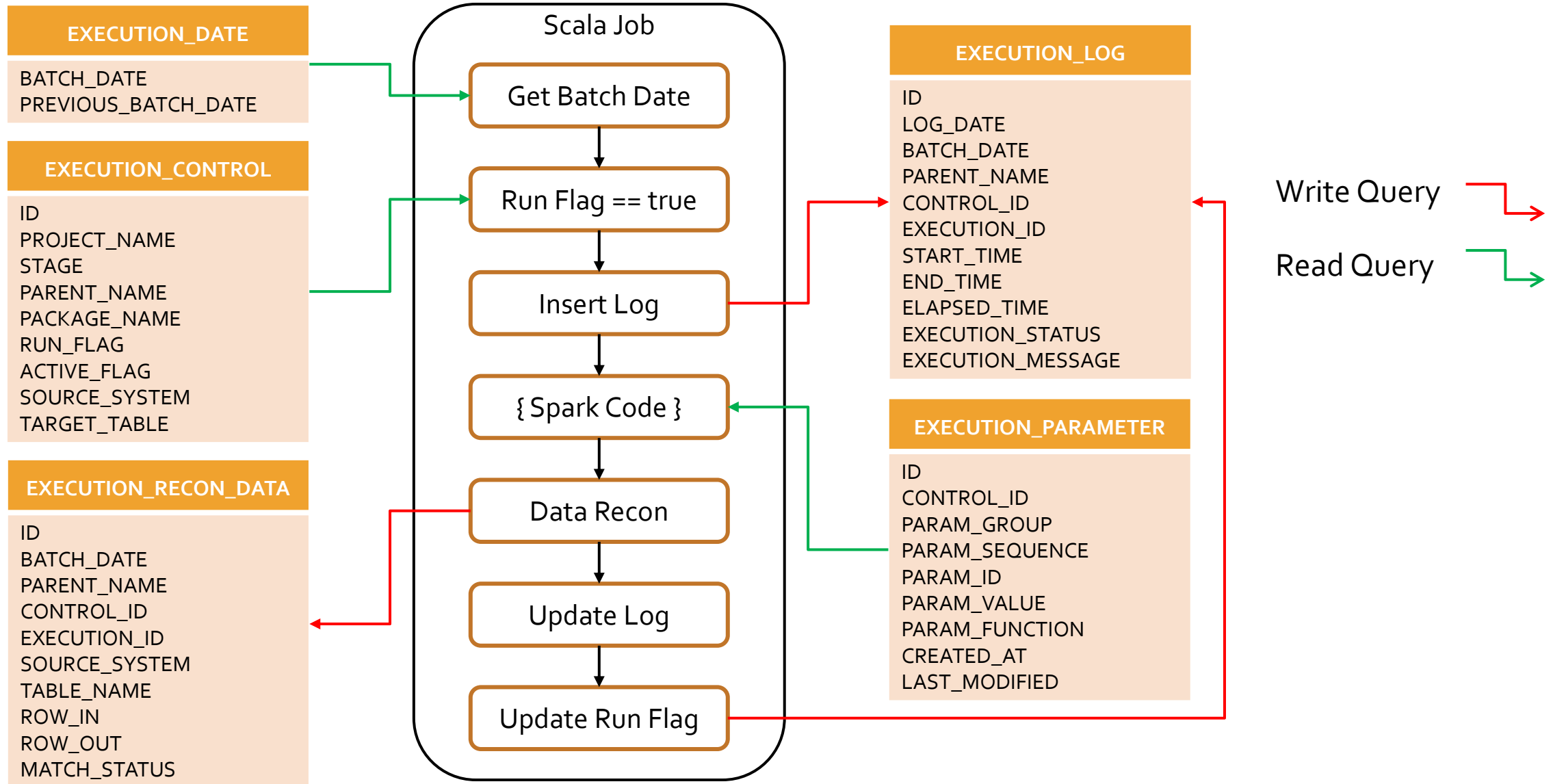
$$is_safe_ratio(min, max) = \frac{|x_2 - x_1|}{x_1} \leq if(x_1 < x_2, max_ratio, min_ratio)$$



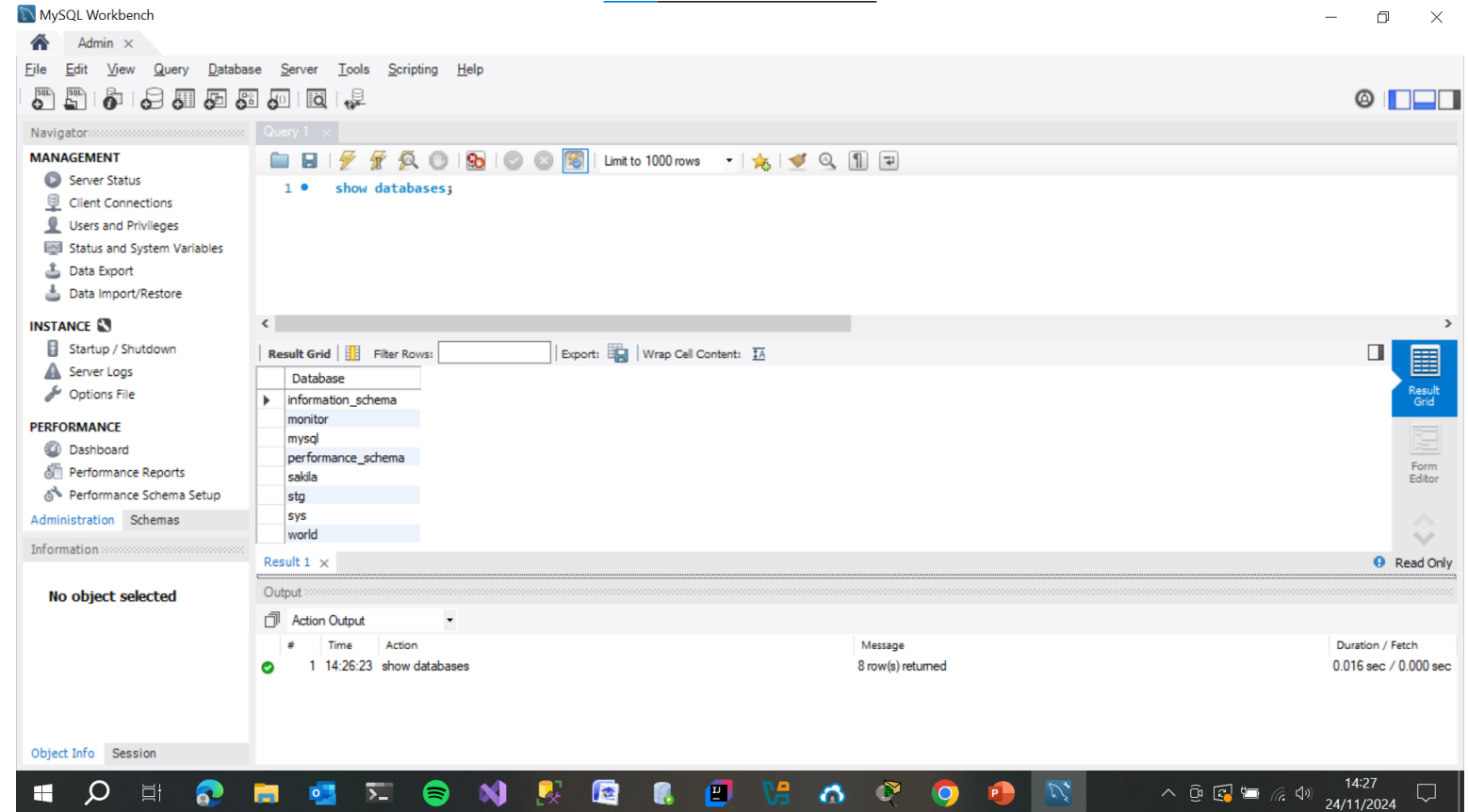
Housekeep Feature



General Scala Spark Framework



Databases



Framework Tables

The screenshot displays the MySQL Workbench interface. The left sidebar contains navigation panels for 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'INSTANCE' (Startup / Shutdown, Server Logs, Options File), and 'PERFORMANCE' (Dashboard, Performance Reports, Performance Schema Setup). The 'Administration' tab is selected, showing 'No object selected'.

The main workspace shows a query window with the following SQL statements:

```
1 show databases;
2 use monitor;
3 show tables;
```

The 'Result Grid' shows the output of the query, with a table named 'Tables_in_monitor' containing the following rows:

Tables_in_monitor
execution_control
execution_date
execution_dependency
execution_failsafe
execution_log
execution_reconciliation

The 'Output' window shows the execution results of the query, with a table containing the following rows:

#	Time	Action	Message	Duration / Fetch
1	14:26:23	show databases	8 row(s) returned	0.016 sec / 0.000 sec
2	14:29:31	use monitor	0 row(s) affected	0.000 sec
3	14:29:40	show tables	6 row(s) returned	0.015 sec / 0.000 sec

The bottom status bar shows the system clock as 14:29 on 24/11/2024.

Execution Control Parameter

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator: Administration - Data Export

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Query 1 x Administration - Data Export

Limit to 1000 rows

```
1 show databases;
2 use monitor;
3 show tables;
4
5 SELECT * FROM execution_control;
```

Result Grid

ID	PROJECT_NAME	STAGE	PARENT_NAME	PACKAGE_NAME	RUN_FLAG	ACTIVE_FLAG	SOURCE_SYSTEM	TARGET_TABLE
1	SPARK-SCALA-TEMPLATE	Template	Template	ChildTemplate	0	1	Template	Template
2	SPARK-SCALA-TEMPLATE	Template	Template	ChildDependencyTemplate	0	1	Template	Template
3	SPARK-SCALA-TEMPLATE	STAGING	MASTER_STG	STG_RENTAL	0	1	sakila.rental	stg.stg_rental
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

execution_control 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	14:26:23	show databases	8 row(s) returned	0.016 sec / 0.000 sec
✓ 2	14:29:31	use monitor	0 row(s) affected	0.000 sec
✓ 3	14:29:40	show tables	6 row(s) returned	0.015 sec / 0.000 sec
✓ 4	18:28:47	SELECT * FROM execution_control LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

Query Completed

18:28 24/11/2024

Execution Date

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator: Administration - Data Export

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Query 1 x Administration - Data Export

Limit to 1000 rows

```
1 show databases;
2 use monitor;
3 show tables;
4
5 SELECT * FROM execution_date;
```

Result Grid

BATCH_DATE	PREVIOUS_BATCH_DATE
2005-05-25	2005-05-24

execution_date 4 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	14:29:31	use monitor	0 row(s) affected	0.000 sec
3	14:29:40	show tables	6 row(s) returned	0.015 sec / 0.000 sec
4	18:28:47	SELECT * FROM execution_control LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
5	18:30:24	SELECT * FROM execution_date LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

18:30 24/11/2024

Execution Log

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator: Administration - Data Export

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Query 1 x Administration - Data Export

```
1 • show databases;
2 • use monitor;
3 • show tables;
4
5 • SELECT `ID`, `LOG_DATE`, `BATCH_DATE`, `PARENT_NAME`, `PACKAGE_NAME`, `START_TIME`,
6   `END_TIME`, `ELAPSED_TIME`, `EXECUTION_STATUS`, `EXECUTION_MESSAGE`
7 FROM execution_log;
```

Result Grid

ID	LOG_DATE	BATCH_DATE	PARENT_NAME	PACKAGE_NAME	START_TIME	END_TIME	ELAPSED_TIME	EXECUTION_STATUS	EXECUTION_MESSAGE
2	2024-11-13	2025-01-02	Template	ChildTemplate	2024-11-13 17:45:44	2024-11-13 17:45:44	0	SUCCESS	NULL
3	2024-11-13	2025-01-02	Template	ChildDependencyTemplate	2024-11-13 20:43:23	2024-11-13 20:43:23	0	SUCCESS	NULL
4	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 22:58:29	2024-11-13 22:58:36	7	ERROR	requirement failed
5	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:01:41	2024-11-13 23:01:45	4	ERROR	requirement failed
6	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:03:17	2024-11-13 23:03:21	4	ERROR	Index 2 out of bounds
7	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:03:44	2024-11-13 23:03:47	3	ERROR	Index 1 out of bounds
8	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:05:25	2024-11-13 23:05:28	3	ERROR	Index 1 out of bounds
9	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:06:04	2024-11-13 23:06:04	0	ERROR	Index 0 out of bounds
10	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:08:12	2024-11-13 23:08:16	4	ERROR	requirement failed
11	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:11:34	2024-11-13 23:11:42	8	SUCCESS	NULL
12	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:12:57	2024-11-13 23:13:05	8	SUCCESS	NULL
13	2024-11-13	2005-05-25	MASTER_STG	STG_RENTAL	2024-11-13 23:14:19	2024-11-13 23:14:27	8	SUCCESS	NULL

Query Completed

18:34 24/11/2024

Execution Reconciliation

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1 x Administration - Data Export

```
1 • show databases;
2 • use monitor;
3 • show tables;
4
5 • SELECT * FROM execution_reconciliation;
```

Limit to 1000 rows

ID	BATCH_DATE	PARENT_NAME	PACKAGE_NAME	EXECUTION_ID	TABLE_NAME	ROW_IN	ROW_OUT	MATCH_STATUS
1	2025-01-01	Template	ChildDependencyTemplate	39653aa0-a1aa-11ef-b192-84a938f0168c	Template	100	100	Y
2	2025-01-02	Template	ChildDependencyTemplate	6f72b1e8-a1ac-11ef-b192-84a938f0168c	Template	85	85	Y
4	2005-05-25	MASTER_STG	STG_RENTAL	f54cb434-a1dc-11ef-b192-84a938f0168c	stg.stg_rental	137	137	1

Result 9 Result 10 execution reconciliation 11 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 11	18:35:32	show databases	8 row(s) returned	0.000 sec / 0.000 sec
✓ 12	18:35:32	use monitor	0 row(s) affected	0.000 sec
✓ 13	18:35:32	show tables	6 row(s) returned	0.000 sec / 0.000 sec
✓ 14	18:35:32	SELECT * FROM execution_reconciliation LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec

Query Completed

18:35 24/11/2024

Execution Dependency

The screenshot shows the MySQL Workbench interface. The left sidebar contains navigation panels for MANAGEMENT, INSTANCE, PERFORMANCE, and Information. The main window displays a query titled "Query 1" with the following SQL statements:

```
1 • show databases;
2 • use monitor;
3 • show tables;
4
5 • SELECT * FROM execution_dependency;
```

The "Result Grid" shows the results of the first three queries:

ID	DEPENDENCY_PACKAGE_NAME	PACKAGE_NAME	MAX_RETRY	DURATION	UNIT_OF_TIME
1	ChildTemplate	ChildDependencyTemplate	3	1	MINUTE
	NULL	NULL	NULL	NULL	NULL

The "Output" panel shows the execution log for the query:

#	Time	Action	Message	Duration / Fetch
✓ 13	18:35:32	show tables	6 row(s) returned	0.000 sec / 0.000 sec
✓ 14	18:35:32	SELECT * FROM execution_reconciliation LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec
✓ 15	18:36:42	show tables	6 row(s) returned	0.016 sec / 0.000 sec
✓ 16	18:37:09	SELECT * FROM execution_dependency LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates "Query Completed" and the system clock shows 18:37 on 24/11/2024.

Execution Failsafe

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator: Administration - Data Export

MANAGEMENT

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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Query 1 x Administration - Data Export

Limit to 1000 rows

```
1 • show databases;
2 • use monitor;
3 • show tables;
4
5 • SELECT * FROM execution_failsafe;
```

Result Grid

ID	PACKAGE_NAME	CHECK_TABLE_NAME	ROWCOUNT_RATIO	CREATED_AT	LAST_MODIFIED
1	ChildDependencyTemplate	Template	20	2024-11-13 17:42:38	2024-11-13 17:42:38

Result 14 Result 15 execution failsafe 16 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 17	18:38:03	show databases	8 row(s) returned	0.000 sec / 0.000 sec
✓ 18	18:38:03	use monitor	0 row(s) affected	0.000 sec
✓ 19	18:38:03	show tables	6 row(s) returned	0.000 sec / 0.000 sec
✓ 20	18:38:03	SELECT * FROM execution_failsafe LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

Query Completed

18:38 24/11/2024

Build.sbt

```
build.sbt x
1  name := "SparkScalaCourse"
2
3  version := "0.1"
4
5  scalaVersion := "2.12.12"
6
7  libraryDependencies ++= Seq(
8    "org.apache.spark" %% "spark-core" % "3.5.3",
9    "org.apache.spark" %% "spark-sql" % "3.5.3",
10   "org.apache.spark" %% "spark-mllib" % "3.5.3",
11   "org.apache.spark" %% "spark-streaming" % "3.5.3",
12   "org.twitter4j" % "twitter4j-core" % "4.0.7",
13   "org.twitter4j" % "twitter4j-stream" % "4.0.7",
14   "mysql" % "mysql-connector-java" % "8.0.33"
15 )
16
```

Result

Spark automatically create table and populate it in distributed manner.

For code please refer the sample.

MySQL Workbench

Admin x

File Edit View Query Database Server Tools Scripting Help

Navigator: Administration - Data Export

MANAGEMENT

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1 x Administration - Data Export

Limit to 1000 rows

```
1 • show databases;
2 • use stg;
3 • show tables;
4
5 • select * from stg_rental;
```

Result Grid

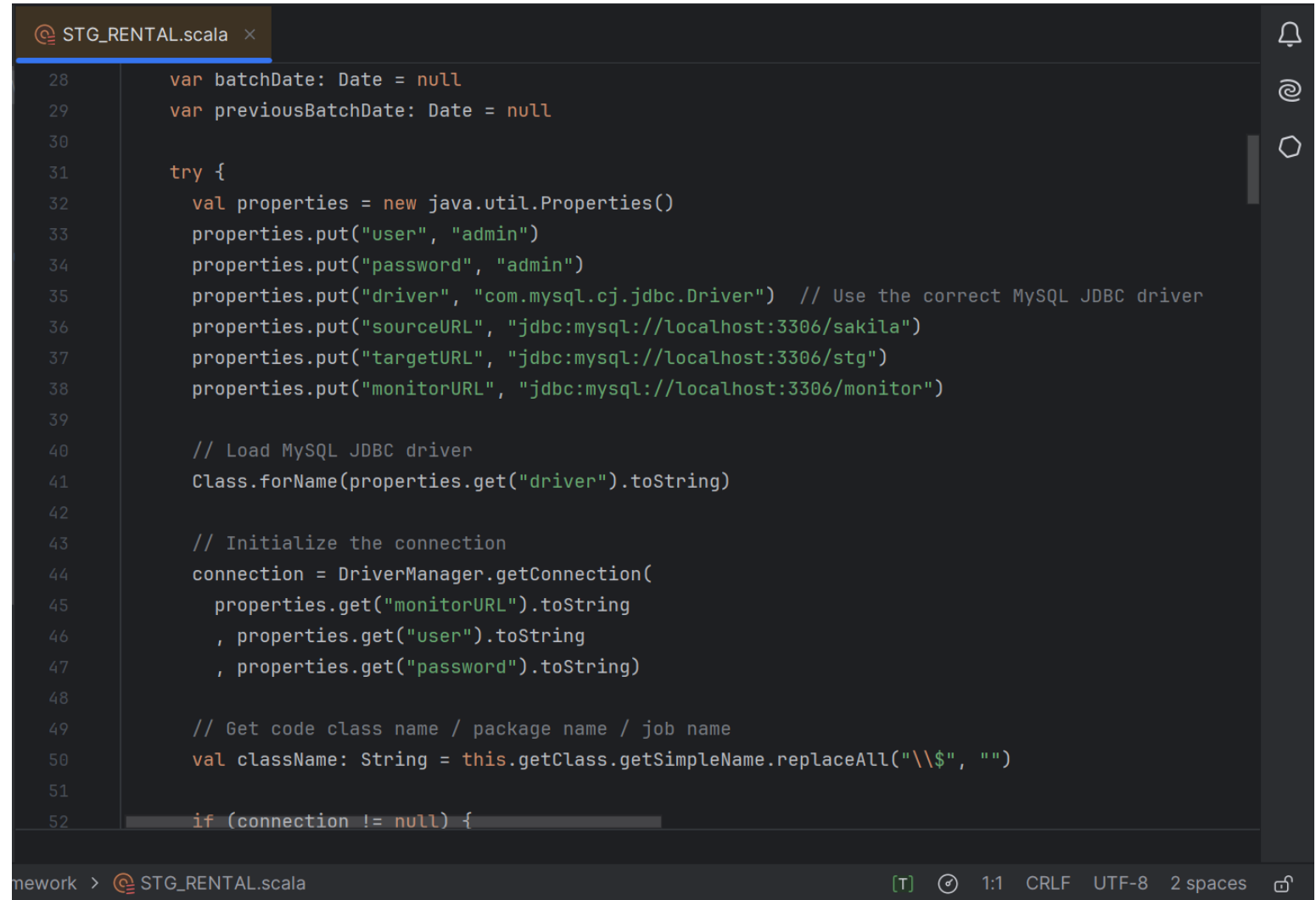
	rental_id	rental_date	inventory_id	customer_id	return_date	staff_id	last_update
9		2005-05-25 00:00:40	2580	126	2005-05-28 00:22:40	1	2006-02-15 21:30:53
10		2005-05-25 00:02:21	1824	399	2005-05-31 22:44:21	2	2006-02-15 21:30:53
11		2005-05-25 00:09:02	4443	142	2005-06-02 20:56:02	2	2006-02-15 21:30:53
12		2005-05-25 00:19:27	1584	261	2005-05-30 05:44:27	2	2006-02-15 21:30:53
13		2005-05-25 00:22:55	2294	334	2005-05-30 04:28:55	1	2006-02-15 21:30:53
14		2005-05-25 00:31:15	2701	446	2005-05-26 02:56:15	1	2006-02-15 21:30:53
15		2005-05-25 00:39:22	3049	319	2005-06-03 03:30:22	1	2006-02-15 21:30:53
16		2005-05-25 00:43:11	389	316	2005-05-26 04:42:11	2	2006-02-15 21:30:53
17		2005-05-25 01:06:36	830	575	2005-05-27 00:43:36	1	2006-02-15 21:30:53
18		2005-05-25 01:10:47	3376	19	2005-05-31 06:35:47	2	2006-02-15 21:30:53
19		2005-05-25 01:17:24	1941	456	2005-05-31 06:00:24	1	2006-02-15 21:30:53
20		2005-05-25 01:48:41	3517	185	2005-05-27 02:20:41	2	2006-02-15 21:30:53
21		2005-05-25 01:59:46	146	388	2005-05-26 01:01:46	2	2006-02-15 21:30:53
22		2005-05-25 02:19:23	727	509	2005-05-26 04:52:23	2	2006-02-15 21:30:53
23		2005-05-25 02:40:21	4441	438	2005-05-29 06:34:21	1	2006-02-15 21:30:53
24		2005-05-25 02:53:02	3273	350	2005-05-27 01:15:02	1	2006-02-15 21:30:53
25		2005-05-25 03:21:20	3961	37	2005-05-27 21:25:20	2	2006-02-15 21:30:53
26		2005-05-25 03:36:50	4171	371	2005-05-31 00:34:50	1	2006-02-15 21:30:53

stg_rental 22 x

Query Completed

18:45 24/11/2024

Open Connection



```
STG_RENTAL.scala x
28     var batchDate: Date = null
29     var previousBatchDate: Date = null
30
31     try {
32         val properties = new java.util.Properties()
33         properties.put("user", "admin")
34         properties.put("password", "admin")
35         properties.put("driver", "com.mysql.cj.jdbc.Driver") // Use the correct MySQL JDBC driver
36         properties.put("sourceURL", "jdbc:mysql://localhost:3306/sakila")
37         properties.put("targetURL", "jdbc:mysql://localhost:3306/stg")
38         properties.put("monitorURL", "jdbc:mysql://localhost:3306/monitor")
39
40         // Load MySQL JDBC driver
41         Class.forName(properties.get("driver").toString)
42
43         // Initialize the connection
44         connection = DriverManager.getConnection(
45             properties.get("monitorURL").toString
46             , properties.get("user").toString
47             , properties.get("password").toString)
48
49         // Get code class name / package name / job name
50         val className: String = this.getClass.getSimpleName.replaceAll("\\$", "")
51
52         if (connection != null) {
```

network > STG_RENTAL.scala [T] 1:1 CRLF UTF-8 2 spaces

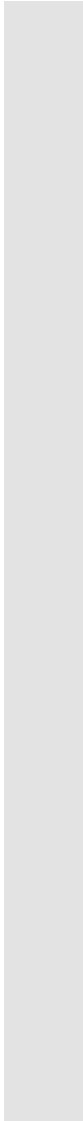
Spark Job Content Example

```
STG_RENTAL.scala x
119 // Create spark session, parallel start here
120 val spark = SparkSession.builder()
121     .appName(className)
122     .master("local[*]")
123     .getOrCreate() // create is previous session is not available
124
125 // Loading data from a JDBC source using load
126 val jdbcDF = spark.read
127     .format("jdbc")
128     .option("url", properties.get("sourceURL").toString)
129     .option("driver", properties.get("driver").toString)
130     .option("dbtable", sourceSystem)
131     .option("user", properties.get("user").toString)
132     .option("password", properties.get("password").toString)
133     .load()
134
135 val filterDF = jdbcDF.filter(to_date(jdbcDF("RENTAL_DATE")) === batchDate)
136
137 filterDF
138     .write
139     .format("jdbc")
140     .mode(SaveMode.Overwrite)
141     .option("url", properties.get("targetURL").toString)
142     .option("driver", properties.get("driver").toString)
143     .option("dbtable", targetTable)
```

network > STG_RENTAL.scala [T] 1:1 CRLF UTF-8 2 spaces

Close Connection

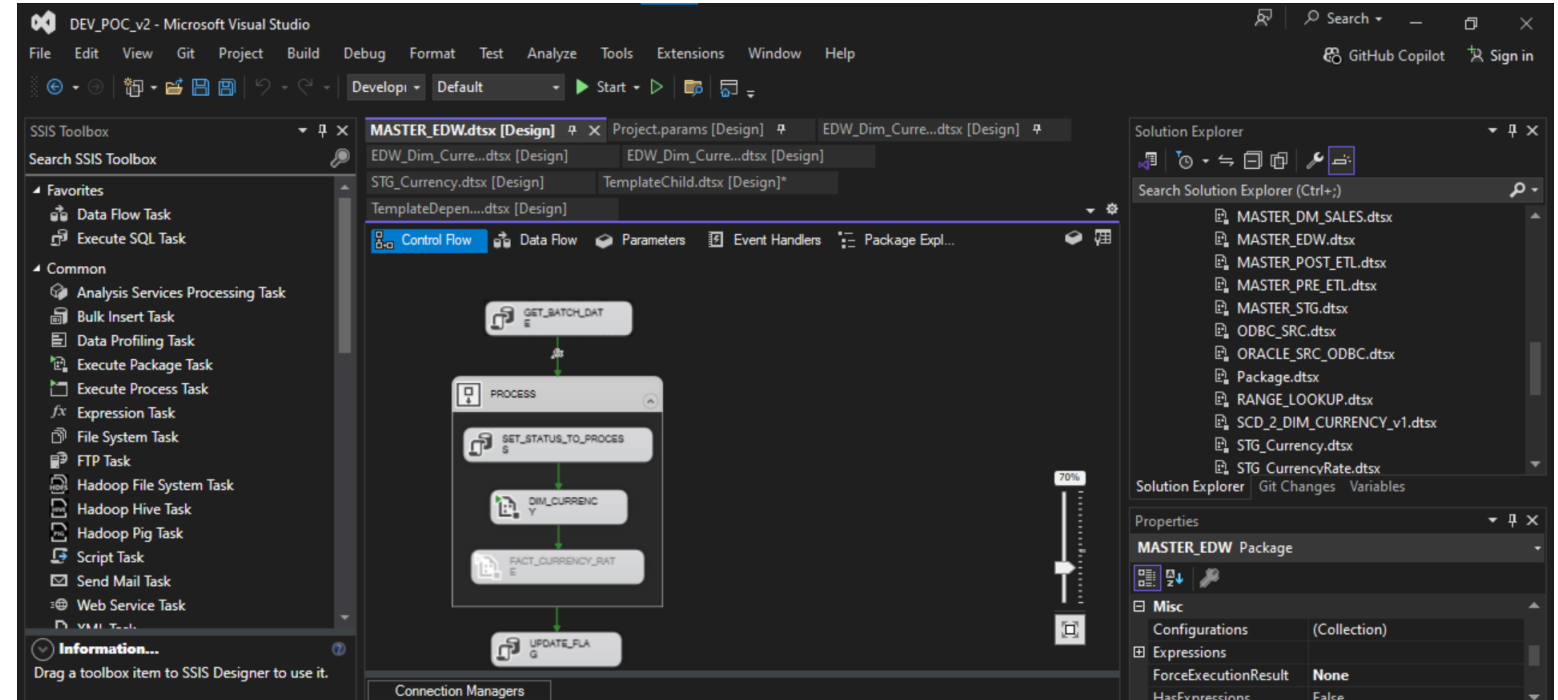
```
STG_RENTAL.scala x
236     }
237     catch {
238       case e: Exception =>
239         // Update log, error status
240         query = "UPDATE `EXECUTION_LOG` SET `END_TIME` = NOW(), " +
241           "`ELAPSED_TIME` = TIMESTAMPDIFF(SECOND, `START_TIME`, NOW()), " +
242           "`EXECUTION_STATUS` = 'ERROR', `EXECUTION_MESSAGE` = ? WHERE `EXECUTION_ID` = ?;"
243
244         preparedStatement = connection.prepareStatement(query)
245         preparedStatement.setString(1, e.getMessage)
246         preparedStatement.setString(2, executionID)
247         preparedStatement.executeUpdate()
248     }
249     finally {
250       println("Closing connection...")
251       try {
252         if (resultSet != null) resultSet.close()
253         if (preparedStatement != null) preparedStatement.close()
254         if (connection != null) connection.close()
255       } catch {
256         case e: SQLException => println("Error closing connection: " + e.getMessage)
257       }
258     }
259   }
260 }
```



Framework Implementation Using Different Tool

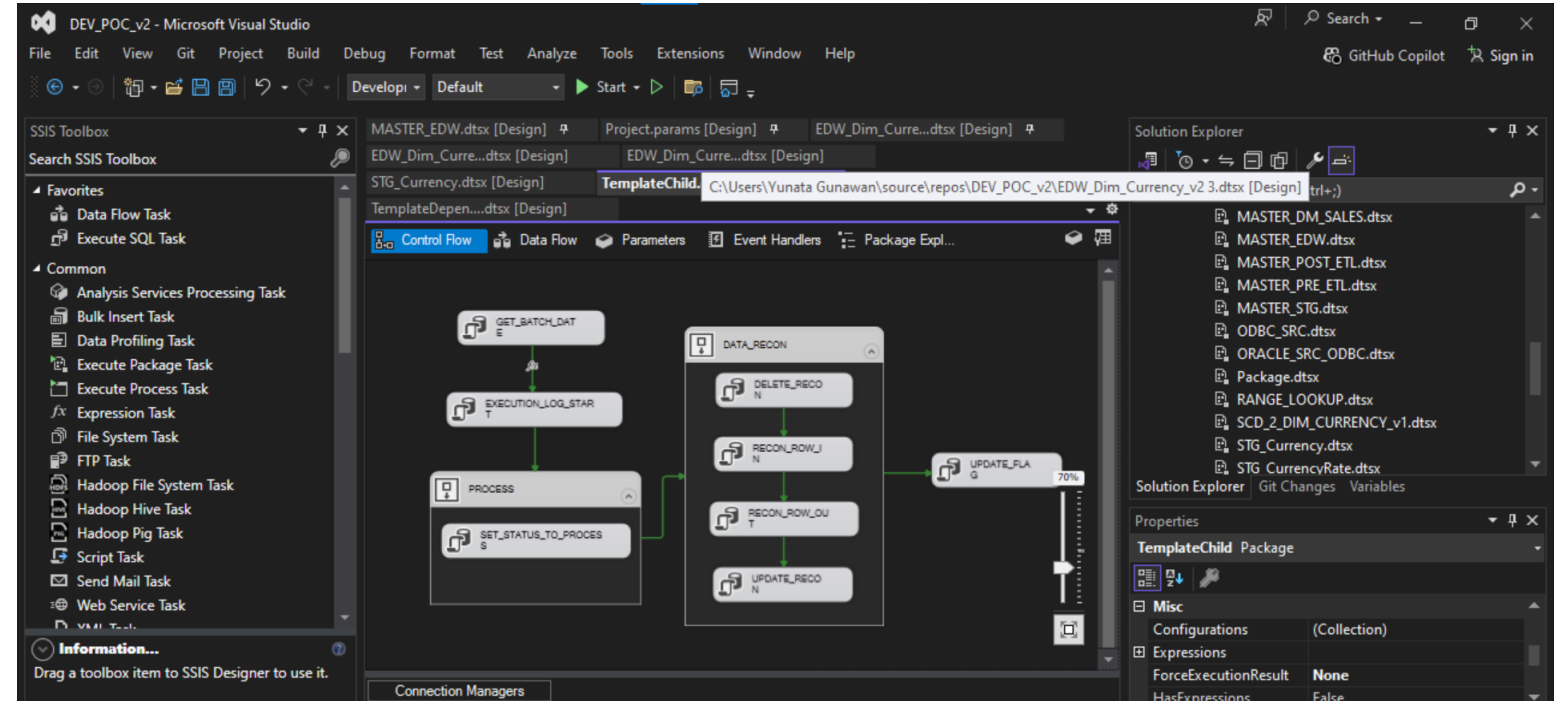
Parent Job Example

Integration Services

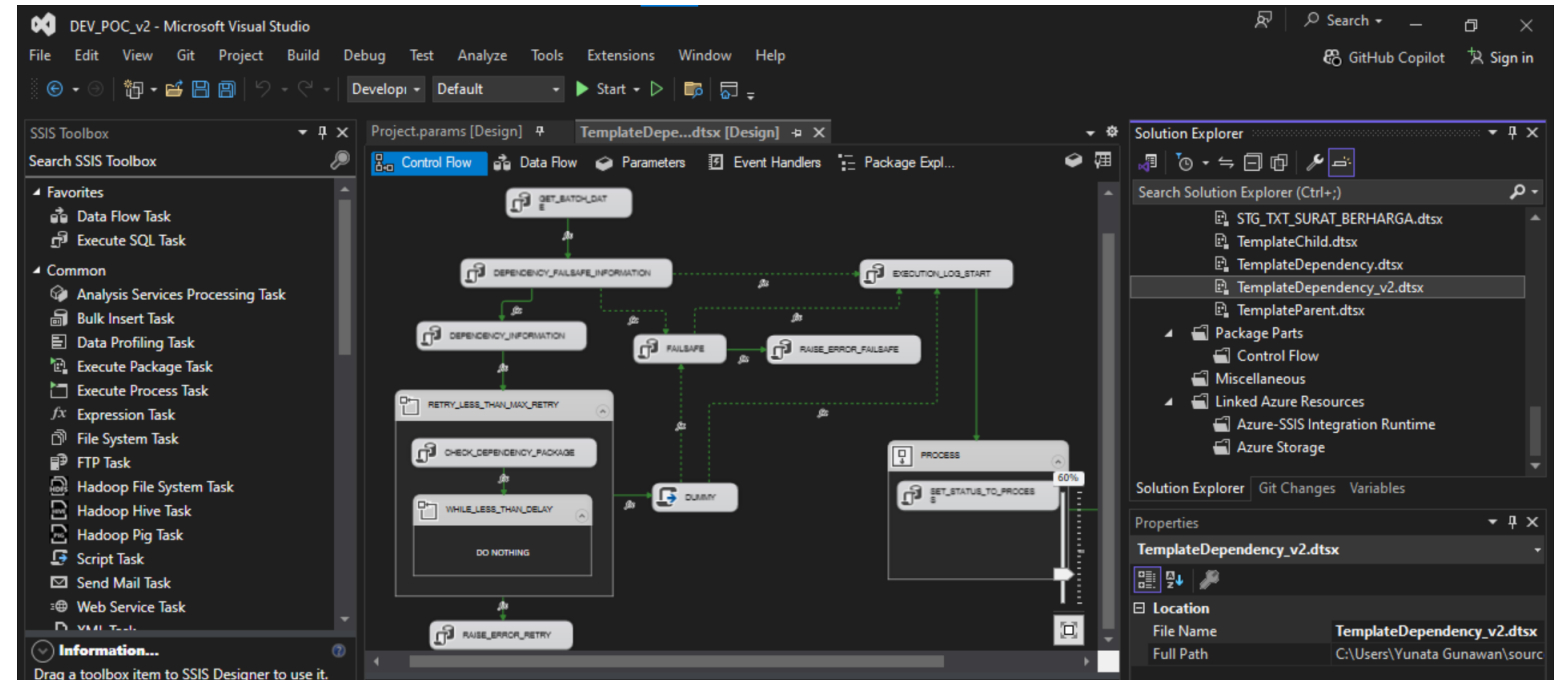


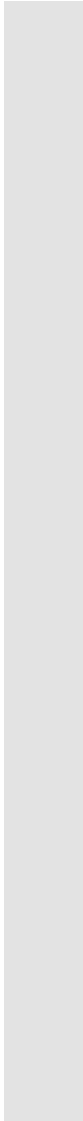
Child Template

Integration Services



Dependency and Failsafe Feature v2





Thank You